

INVESTIGATION OF SUSPENSION AND STEERING

PERFORMANCE OF THE MCLAREN F1

Abstract

Due to major competition in the automotive industries these days everyone is looking for the best car to manufacture sell and buy. To achieve this task the companies and the customer actively seek the full car specification of their rival's product with their own product for comparison and improving for a better model. Many automotive companies only give general specifications like maximum speed, general dimensions and suspension and steering specification, but as an engineer we need to seek and investigate the full specification of a car to a point where we can reverse engineer the car using these data.

Introduction and background

The "fastest car of the world" is always a fascinating title. Lamborghini, Ferrari, Porsche and Bugatti all chased this title during the 60's to 80's. In 1993, McLaren launched McLaren F1, and finished this competition with the 231 mph highest speed powered by BMW S70/2 6.1 litre V12 engine. McLaren F1 was world's fastest car as well as the world's fastest production car in 1998. The chassis body of McLaren F1 is carbon fibre reinforced composite monocoque, it sets a precedent for carbon fibre to lightweight. The design of McLaren are very advanced at that time and many designs are still in use of many formula car today. McLaren F1 is still known as the best sportscar of all time.

This report calculated many parameters of McLaren F1, including the load distribution, kinetic and static tyre load, wheel and tyre specification and other related aspects using the knowledge gained through our course. By calculate the various data of McLaren F1, aim to seek out the suspension and steering system characteristic and get a fully understand of car specification in order to follow the process of reverse engineering the car. Reverse engineering is a process which many companies use to recreate the product using the data gained for comparing with their own product to improve it. The design and of McLaren are so advanced that many automotive companies still seek its data to improve their current model of sports cars.

Technical content

Calculating the tyre stiffness

1) Weight of front=1140*9.81*0.42=4697 N

2349 N/one front wheel

Load on tyre=Pressure*Area (contact patch) =Pressure*Length*width; assume that width will not change.

So, 2349 N=2.3*10⁵*length*235 mm

So, length front = 43.5 mm

2) Weight of rear =1140*9.81*0.58 = 6486 N

3243 N/one rear wheel

3243=2.8*10⁵*Length*315 mm

So, length rear = 36.8 mm

3) Front tyre Ø= (2*0.45*235) + (17*25.4) =643.3mm

Radius = 321.6mm

Rear tyre Ø= (2*0.45*315) + (17*25.4) =715.3mm

Radius=357.7mm

4) Tyre Deflection $\delta = R - h$

$$h = \sqrt{R^2 - \left(\frac{\text{length}}{2}\right)^2}$$

h front=320.9 mm

h rear=356 mm

$\delta_{\text{front}} = 321.6 - 320.9 = 0.7\text{mm}$

$\delta_{\text{rear}} = 357.7 - 357.2 = 0.5\text{mm}$

So, tyre stiffness:

front= $\frac{\text{load}}{\delta} = \frac{2349}{0.7}=3356\text{N/mm}$

rear= $\frac{3243}{0.5} = 6486\text{N/mm}$

Parameter	Value	Source
Curb weight	1140 kg	Given from vehicle data
Weight at front	42%	Given from vehicle data
Weight at rear	58%	Given from vehicle data
Pressure of tyre front	2.3 bar	Assumed since it is the standard pressure for 235/45ZR17 tyre for road cars

Pressure of tyre rear	2.8 bar	Assumed since it is the standard pressure for 315/45ZR17 tyre for road cars
Radius of front tyre	321.6mm	calculated
Radius of rear tyre	357.7mm	calculated
Load on front tyre	2349N	calculated
Load on rear tyre	3243N	calculated
Tyre stiffness on front	3356 N/mm	calculated
Tyre stiffness on rear	6486 N/mm	calculated

Determine the spring stiffness of the car and the number of active coils.

1) Bounced frequency at front: 1.43Hz.

$$\omega = 2\pi f = 2\pi * 1.43 = 8.986 \text{ rad/s}$$

$$\text{Rear } f=1.8\text{Hz}, \omega = 2\pi f = 2\pi * 1.8 = 11.31 \text{ rad/s}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\text{so, } K = \omega^2 m$$

$$2) m_f = \left(\frac{1140 * 0.42}{2} \right) - 40 = 199 \text{ kg}$$

$$m_r = \left(\frac{1140 * 0.58}{2} \right) - 50 = 280.6 \text{ kg}$$

$$3) k_{sf} = \frac{199 * 8.986^2}{1000} = 16.1 \text{ N/mm}$$

$$k_{sr} = \frac{280.6 * 11.31^2}{1000} = 35.89 \text{ N/mm}$$

$$4) \text{ spring rate } k_{\text{spring}} = \frac{Gd^4}{8nD^3}$$

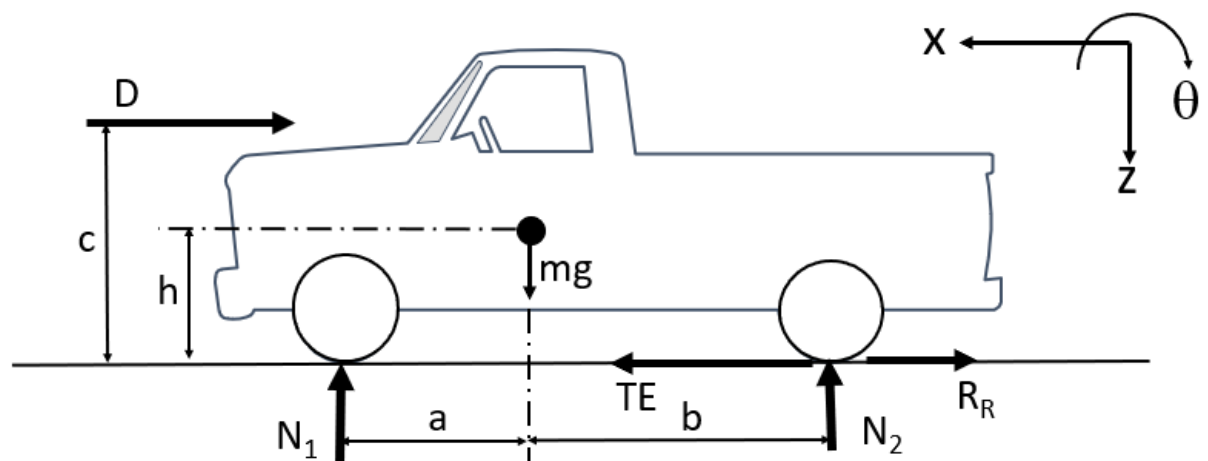
$$n = \frac{Gd^4}{8k_{\text{spring}}D^3} = \frac{79.3 \times 10^3 \times 12.7^4}{8 \times 16.1 \times 120^3} = 9.27 \text{ front coils}$$

$$n = \frac{Gd^4}{8k_{\text{spring}}D^3} = \frac{79.3 \times 10^3 \times 12.7^4}{8 \times 35.89 \times 100^3} = 7.18 \text{ rear coils}$$

Parameter	Value	Source
Bounced frequency at front	1.43 Hz	Given from vehicle data
Bounced frequency at rear	1.80 Hz	Given from vehicle data
Unsprung mass front corner	40 kg	Searching from internet and

		assume
Unsprang mass rear	50 kg	Searching from internet and assume
Wire diameter	12.7 mm	Searching from internet and assume
Front mean coil diameter	120 mm	Searching from internet and assume
Rear mean coil diameter	100 mm	Searching from internet and assume
Spring shear modulus	79.3 GN/m ²	Searching from internet and assume
Spring stiffness at front	16.1 N/mm	calculated
Spring stiffness at rear	35.89 N/mm	calculated
Active coils at front	9.27	calculated
Active coils at rear	7.18	calculated

Load Transfer effects and individual tyre loading during steady state conditions, acceleration and braking.



The acceleration and deceleration of the car is based on the follow scenario:

Acceleration: 0-70mkp $a=7\text{m/s}^2$

Deceleration: 70-0mkp $a=9.8\text{m/s}^2$

1) $b=2718 \times 0.42=1141.56 \text{ mm}$

$a=2718 \times 0.58=1576.44 \text{ mm}$

2) Aerodynamic

$$D = \frac{\rho c_d A v^2}{2} = \frac{1.225 * 0.32 * 1.79 * (112/3.6)^2}{2} = 340 \text{ N}$$

3) Assume c=0.5 m

The tyres — 235/45ZR17 (643.3 dia) front and 315/45ZR17 rear (715.3 dia),

The disc sizes are 332 front and 305 rear

The pad radial depth is about 40mm

4) Front disc effective diameter 332-40 = 292mm

rear disc effective diameter 305 – 40 = 265mm

Front braking force = disc size/tyre size = 292/643.3 = 0.454

Rear braking force = disc size/tyre size = 265/715.3 = 0.370

Front proportion = 0.454/ (0.454+0.370) = 55.1%

Rear proportion = 0.370/ (0.454+0.370) = 44.9%

A 13.1% load transfer

$$h = 0.131 * 1140 * 9.81 * 2.718 / (1140 * 9.81) = 347 \text{ mm}$$

Steady State:

$$\text{Load } N_{1L} = N_{1R} = \frac{mgb - Dc}{2(a+b)} = \frac{1140 * 9.81 * 1.142 - 340 * 0.5}{2.718 * 2} = 2318 \text{ N}$$

$$\text{Load } N_{2L} = N_{2R} = \frac{mga + Dc}{2(a+b)} = \frac{1140 * 9.81 * 1.576 + 340 * 0.5}{2.718 * 2} = 3274 \text{ N}$$

While Acceleration:

$$\text{Load } N_{1L} = N_{1R} = \frac{mgb - Dc - mxh}{2(a+b)} = \frac{1140 * 9.81 * 1.142 - 340 * 0.5 - 1140 * 7 * 0.347}{2 * 2.718} = 1808.75 \text{ N}$$

$$\text{Load } N_{2L} = N_{2R} = \frac{mga + Dc + mxh}{2(a+b)} = \frac{1140 * 9.81 * 1.576 + 340 * 0.5 + 1140 * 7 * 0.347}{2 * 2.718} = 3782.95 \text{ N}$$

While Braking:

$$\text{Load } N_{1L} = N_{1R} = \frac{mgb - Dc + mxh}{2(a+b)} = \frac{1140 * 9.81 * 1.142 - 340 * 0.5 + 1140 * 9.8 * 0.347}{2 * 2.718} = 3031.30 \text{ N}$$

$$\text{Load } N_{2L} = N_{2R} = \frac{mga + Dc - mxh}{2(a+b)} = \frac{1140 * 9.81 * 1.576 + 340 * 0.5 - 1140 * 9.8 * 0.347}{2 * 2.718} = 2560.40 \text{ N}$$

Parameter	Value	Source
Drag coefficient cd	0.32	Given from vehicle data
Front area A	1.79 m ²	Given from vehicle data
Velocity v	112 km/h (70 mph)	Given from vehicle data
Drag force D	340 N	calculated
Drag force height c	0.5 m	Searching from internet and assume
Height of Cog h	347 mm	calculated

Load on front wheel (steady)	2318 N	calculated
Load on rear wheel (steady)	3274 N	calculated
Load on front wheel (accelerating)	1808.75 N	calculated
Load on rear wheel (accelerating)	3782.95 N	calculated
Load on front wheel (braking)	3031.30 N	calculated
Load on rear wheel (braking)	2560.40 N	calculated

Estimate friction demands for each tyre during straight driving and cornering.

1) Assume the overall transmission efficiency is 0.9

$$\text{Tractive effort } TE_f = \frac{T_e n_g n_d \eta}{r_r} = \frac{651 \cdot 3.23 \cdot 2.37 \cdot 0.9 \cdot 1000}{357} = 12563.37 \text{ N}$$

2) Straight driving

For rear tyre friction demands = 12563.37N

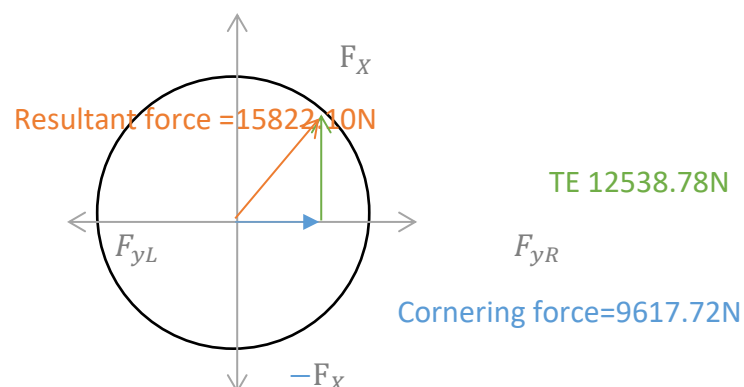
2) Cornering driving

For steady state cornering the friction demand side to side remains the same for each axle

Cornering force = $ma = 0.86 \cdot g \cdot m = 9617.724 \text{ N}$

For rear tyre resultant force = $\sqrt{F_x^2 + F_y^2} = \sqrt{12563.37^2 + 9617.72^2} = 15822.10 \text{ N}$

Friction circle:



parameter		value	source
T_e	Torque at engine	651 Nm	given
n_g	Gearbox ratio	3.23:1	given
n_d	Differential ratio	2.37:1	given
η	Overall efficiency	0.9	assumption
TE	Tractive effort	13246.17N	calculation
$F_{cornering}$	Cornering force	9617.72N	calculation
$F_{resultant}$	Resultant force	15801.96N	calculation

Range of accelerations (test by autocar magazine)

change of velocity	Time(s)	Acceleration(m/s^2)
0–30 mph (13.33m/s)	1.8	7.41
0–40 mph (17.78m/s)	2.3	7.73
0–50 mph (22.22m/s)	2.7	8.23
0–60 mph (26.94 m/s)	3.2	8.42
0–70 mph (31.39 m/s)	3.9	8.05
0–100 mph (44.72m/s)	6.3	7.10
0–124.3 mph (55.56/s)	9.4	5.91
0–150 mph (66.94m/s)	12.8	5.23
0–200 mph (88.89m/s)	28	3.17
30–50 mph (13.33–22.22m/s)	1.8	4.94
30–70 mph (13.33–31.39m/s)	2.1	8.60
40–60 mph (17.78–26.94 m/s)	2.3	3.98
50–70 mph (22.22–31.39 m/s)	2.8	3.27
180–200 mph (80.56–89.44 m/s)	7.6	1.17

The range of the acceleration is from $1.17 m/s^2$ to $8.42 m/s^2$.

During heavy acceleration, the steering system maybe lost control. That's because of when the vehicle is acceleration, the load in font wheel is decreasing due to the load transfer effect.

Tyre loads during cornering, acceleration and braking

1) Centrifugal force (CF) = $ma = 0.86 \times g \times 1140 = 9617.724N$

Given Front track = 1568 mm

Given Rear track = 1472 mm

2) Cornering tyre load = static $\pm LTE$

Based on the previous calculation

$$\text{Static Load } N_{fL} = N_{fR} = \frac{mgb - Dc}{2(a+b)} = 2318 \text{ N}$$

$$N_{rL} = N_{rR} = \frac{mga + Dc}{2(a+b)} = 3274 \text{ N}$$

$$b = 2718 * 0.42 = 1141.56 \text{ mm}$$

$$a = 2718 * 0.58 = 1576.44 \text{ mm}$$

$$LTE_f = CF \left(\frac{b}{(a+b)} \right) * \frac{h}{T_f} = 9617.724 * \left(\frac{1142}{2718} \right) * \frac{347}{1568} = 894.28 \text{ N}$$

$$LTE_r = CF \left(\frac{a}{(a+b)} \right) * \frac{h}{T_r} = 9617.724 * \left(\frac{1576}{2718} \right) * \frac{347}{1472} = 1314.62 \text{ N}$$

$$\text{For front left tyre } N1 = N_{fL} - LTE_f = 2318 - 894.28 = 1423.72 \text{ N}$$

$$\text{For front right tyre } N2 = N_{fR} + LTE_f = 2318 + 894.28 = 3212.28 \text{ N}$$

$$\text{For rear left tyre } N3 = N_{rL} - LTE_r = 3274 - 1314.62 = 1959.38 \text{ N}$$

$$\text{For rear right tyre } N4 = N_{rR} + LTE_r = 3274 + 1314.62 = 4588.62 \text{ N}$$

3) Acceleration

$$LTE = \frac{mah}{wheelbase} = \frac{1140 * 7 * 347}{2718} = 1223.30 \text{ N}$$

$$\text{For front left tyre } N1 = N_{fL} - LTE_f - LTE = 2318 - 894.28 - 1223.30 = 200.42 \text{ N}$$

$$\text{For front right tyre } N2 = N_{fR} + LTE_f - LTE = 2318 + 894.28 - 1223.30 = 1988.98 \text{ N}$$

$$\text{For rear left tyre } N3 = N_{rL} - LTE_r + LTE = 3274 - 1314.62 + 1223.30 = 3182.68 \text{ N}$$

$$\text{For rear right tyre } N4 = N_{rR} + LTE_r + LTE = 3274 + 1314.62 + 1223.30 = 5811.92 \text{ N}$$

4) Breaking

$$LTE = \frac{mah}{wheelbase} = \frac{1140 * 9.8 * 347}{2718} = 1426.30 \text{ N}$$

$$\text{For front left tyre } N1 = N_{fL} - LTE_f + LTE = 2318 - 894.28 + 1426.30 = 2850.02 \text{ N}$$

$$\text{For front right tyre } N2 = N_{fR} + LTE_f + LTE = 2318 + 894.28 + 1426.30 = 4638.58 \text{ N}$$

$$\text{For rear left tyre } N3 = N_{rL} - LTE_r - LTE = 3274 - 1314.62 - 1426.30 = 533.08 \text{ N}$$

$$\text{For rear right tyre } N4 = N_{rR} + LTE_r - LTE = 3274 + 1314.62 - 1426.30 = 3162.32 \text{ N}$$

L	Wheelbase	2718mm	given
T_f	Front track	1568mm	given
T_r	Rear track	1472mm	given

CF	Centrifugal force	9617.72 N	calculation
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Result

	Cornering(N)	Acceleration(N)	Deceleration(N)
Front left tyre	1423.72	200.42	2850.02
Front right tyre	3212.28	1988.98	4638.58
Rear left tyre	1959.38	3182.68	533.08
Rear right tyre	4588.62	5811.92	3162.32

The limits of the tyre /road friction levels(adhesion)when downforce is considered

When the vehicle starts to move, $ma =$, so $\mu = \frac{ma}{N} = \frac{1138 \times 8.42}{1138 \times 9.81} = 0.85$

Assume the lift coefficient is 0.8, when the car is in highest velocity

$$F_{down} = \frac{1}{2} * WS * H * f * \rho * v^2 = \frac{1}{2} * 1.82 * 1.14 * 0.8 * 1.23 * \left(\frac{371.68}{3.6}\right)^2 = 10899.89N$$

$$\text{Front wheel load } F_1 = \mu(N_r + \frac{F_{down}}{2}) = 0.85 * (4688.78 + 10899/2) = 8617.54N$$

$$\text{Tyre load} = 4308.77N$$

$$\text{Rear wheel load } F_2 = \mu(N_f + \frac{F_{down}}{2}) = 0.85 * (6474.99 + 10899/2) = 10135.82N$$

$$\text{Tyre load} = 5067.91N$$

Steering geometry and Ackermann's angles

Formulas Used

$$\tan \delta_i = \frac{L}{R - \frac{T}{2}}$$

$$\tan \delta_{OA} = \frac{L}{R + \frac{T}{2}}$$

$$R = \frac{L}{\tan \delta_{OA}} - \frac{T}{2}$$

$$\sin \delta_{OA} = \frac{L}{D_s/2 - d}$$

$$D_s = 2 \left(\frac{L}{\sin \delta_{OA}} + d \right) + \frac{\text{tyre width}}{2}$$

Where,

L - Wheelbase of the vehicle.

R - Mean radius of turn at rear axle.

T^* - Track- The distance between 2 centres of a wheel in an axel.

$T = T^* - 2d$

δ_{OA} - Outer wheel angle.

δ_i - Inner wheel angle.

D_s - Turning circle diameter.

d - Lateral Inclination Offset

From the data sheet we know,

$L = 2718\text{mm}$

$d = 16.25\text{mm}$ (Martin, 2014)

$T^* = 1472\text{mm}$

$T = T^* - 2d = 1472 - 2(16.25) = 1439.5$

$D_s = 13000\text{mm}$

From the given data we can calculate the Outer Wheel Angle,

$$\sin \delta_{OA} = \frac{L}{D_s/2 - d}$$

$$\sin \delta_{OA} = \frac{2718}{13000/2 - 16.25} = 0.419201$$

$$\delta_{OA} = \sin^{-1} 0.419201 = 24.784^\circ$$

Using Outer Wheel Angle, we can determine Mean radius of turn,

$$\tan \delta_{OA} = \frac{L}{R + \frac{T}{2}}$$

$$R = \frac{L}{\tan \delta_{OA}} - \frac{T}{2}$$

$$R = \frac{2718}{\tan 24.784} - \frac{1439.5}{2} = 5166.8016\text{mm}$$

Therefore, the Inner Wheel angle is,

$$\tan \delta_i = \frac{L}{R - \frac{T}{2}}$$

$$\tan \delta_i = \frac{2718}{5166.8016 - \frac{1439.5}{2}} = 0.6111$$

$$\delta_i = \tan^{-1} 0.6111 = 31.4329^\circ$$

Differential steer angle or toe difference angle,

$$\delta_A = \delta_i - \delta_{OA} = 31.4329 - 24.7842 = 6.6487^\circ$$

Curb to Curb theoretical turning diameter

$$D_S = 2 \left(\frac{L}{\sin \delta_{OA}} + d \right) + \frac{\text{tyre width}}{2}$$

$$D_S = 2 \left(\frac{2718}{0.419201} + 16.25 \right) + \frac{235}{2} = 13235.05725 \text{ mm}$$

Tyre scrub

Formulas Used

$$A = \frac{\pi D^2}{4}$$

$$D = \sqrt{\frac{4A}{\pi}}$$

$$D = a$$

$$\frac{d}{a}$$

$$h = \sqrt{d^2 + \left(\frac{a}{3}\right)^2}$$

$$M_S = \mu_e w h$$

Where,

A- Contact Patch area

D=a- Contact patch Diameter

h- Effective moment arm

μ_e - Coefficient of Friction

W- Vertical load

M_S - Tyre scrub torque or Turning Scrub Moment

From the Data sheet we know,

$$A_f = 0.01021 \text{ m}^2$$

$$A_r = 0.11582 \text{m}^2$$

$$d = 16.25 \text{mm}$$

Area of Contact Patch

$$A = \frac{\pi D^2}{4}$$

$$D = a_f = \sqrt{\frac{4A_f}{\pi}} = \sqrt{\frac{4(0.01021)}{\pi}} = 0.11401 \text{m} = 114.016 \text{mm}$$

$$D = a_r = \sqrt{\frac{4A_r}{\pi}} = \sqrt{\frac{4(0.011582)}{\pi}} = 0.121435 \text{m} = 121.435 \text{mm}$$

Ratio d/a

$$\frac{d}{a_f} = \frac{16.25}{114.016} = 0.142$$

$$\frac{d}{a_r} = \frac{16.25}{121.435} = 0.1338$$

Using the Empirical curves for coefficient of friction curve we can determine coefficient of friction

$$\mu_e = 0.5 \text{ (as for both front and rear tyres the ratio } d/a \text{ is almost equal)}$$

Therefore, we can calculate effective moment arm using,

$$h_f = \sqrt{d^2 + (a_f/3)^2} = \sqrt{16.25^2 + (114.016/3)^2} = 41.333 \text{mm}$$

$$h_r = \sqrt{d^2 + (a_r/3)^2} = \sqrt{16.25^2 + (121.435/3)^2} = 43.618 \text{mm}$$

Hence, the Tyre Scrub Torque during,

$$M_{Sf} = \mu_e w h_f = 0.5 \times 2349 \times 41.333 = 48545.608 \text{Nmm}$$

$$M_{Sr} = \mu_e w h_r = 0.5 \times 3243 \times 43.618 = 70726.587 \text{Nmm}$$

Induced torque during Acceleration

$$M_{Sf} = \mu_e w h_f = 0.5 \times 1808.75 \times 41.333 = 37380.531 \text{Nmm}$$

$$M_{Sr} = \mu_e w h_r = 0.5 \times 3782.95 \times 43.618 = 82502.35 \text{Nmm}$$

Induced torque during Braking

$$M_{Sf} = \mu_e w h_f = 0.5 \times 3031.30 \times 41.333 = 62646.361 Nmm$$

$$M_{Sr} = \mu_e w h_r = 0.5 \times 2560.40 \times 43.618 = 55839.763 Nmm$$

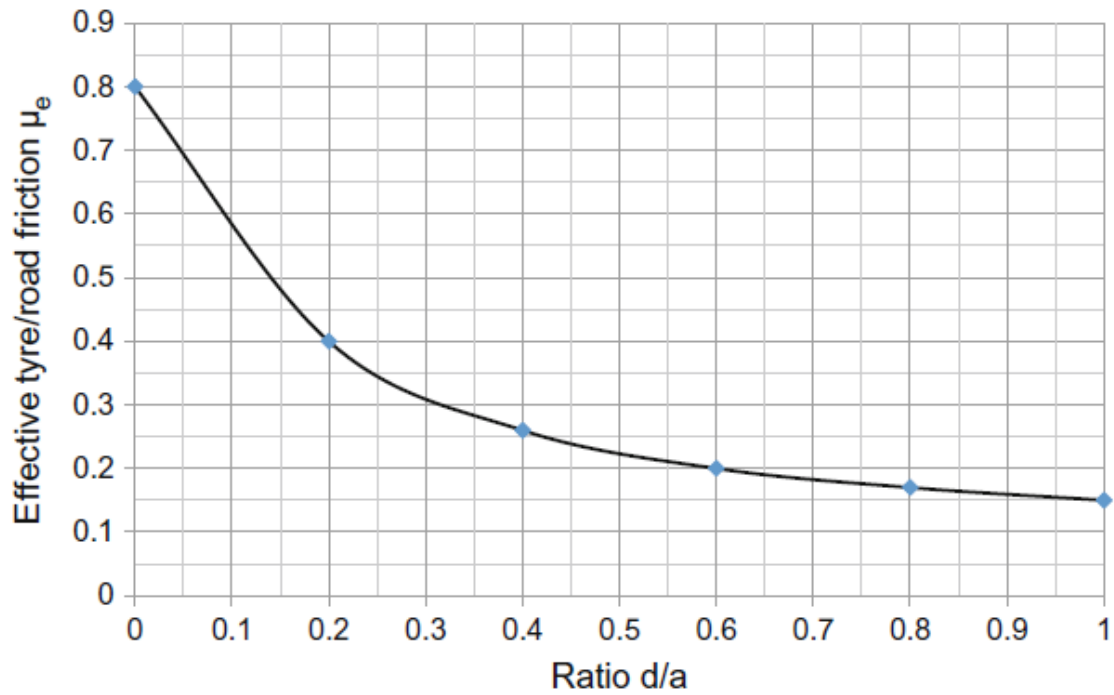


Figure 1 Empirical curves for coefficient of friction

Discussion and conclusions

From the above technical data, it can be said that McLaren F1 is of hardest car specification to investigate. Most of the general specification given isn't enough to calculate the full specification of a car for reverse engineering. Most data had to be assumed as the standard data for any road legal sports car as they weren't available in any sources and journals' can also assume that McLaren F1 has a complex design as it took 6 years even for McLaren to design and finish the production of this car. It was such a great car that it took 21 year for McLaren to produce a supercar close to McLaren F1.

From the full data gathered above we can determine that the car can accelerate at $7m/s^2$ to decelerate at $9.8m/s^2$. The aerodynamics of the car design is also great comparatively to the other cars of Mercedes, jaguar, Aston martin etc., as the aerodynamic drag force acting on the car is 340N which is good for cornering at high speeds. The tyre selected for the car is

also perfect for handling its load and speed as most tyre with less stiffness might explode at high speeds. Every object has its own frequency and the best condition for us to travel in a car is when the natural frequency is in the range of 1Hz to 1.5Hz. the passengers will feel uncomfortable if the frequency increases to 2Hz. The Bounced frequency of McLaren 1.43Hz which is idea for us to travel in it without any discomfort.

The suspension is a double wishbone system with unusual design which retract the wheels backwards increasing its comfort when hitting a bump. The data for the suspension suggest that it is good enough for handle cornering and also harsh road conditions without causing any problems for the car and its passengers. During acceleration there is a chance of steering system to lose control as the load in front wheel is decreasing due to the load transfer effect. It has large turning circle just like many other sports cars of its time and the tyre scrub torque is similar compared with the other sportscars. The reduced torque during acceleration and deceleration with less drag can signify that the fuel consumption is less and it is economical when comes to driving the car. The steering system is simple rack and pinion type geometry which is harder to steer compared to the modern days hydraulic system but simpler and easier to avoid any unnecessary issues. It takes two and half turns of steering from lock to lock of the tyre which is good as it requires less effort to turn the car. The carbon fibre composite chassis is like a prototype for the formula one cars of McLaren team. As it reduces the weight significantly unlike Jaguar XJ220 or Testarossa, both of which weigh hundreds of kilos more than F1. Since it is a 3-seater and has a central driving system just like formula one cars the driver is safe in case of roll or side impact of the car. From the above calculated data, we can say that McLaren has done a great job designing this car as it is really efficient and economical and its design was way ahead of its time of production and will always be the best car ever produced.

In conclusion we can say that McLaren F1 in a great car and still the best sports car ever made which is not just about speed but also about efficiency and life expectancy. But most of the data gathered and calculated might not be accurate as per the real model of the car. The values might be more accurate if more specification can be given especially for the suspension and steering system as they had the least available data. So, a request for more experimental and geometric data is needed for reverse engineering of McLaren F1.

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Martin, W. (2014). Suspension. *Projects, Tilting Three Wheeler*.